

Appendix A: Basic Algorithm Information

Algorithm Name	LoongKEM	
Algorithm Functionality (Single choice)	☐ Digital Signature ☒ Key Encapsulation ☐ Key Exchange	
Submitter 1 <i>Chengdong Tao</i> 2026-08-30	Name	Chengdong Tao
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Appendix B: Intellectual Property Statements

B.1 Statement by Each Algorithm Submitter

I, Chengdong Tao, Room 2312, Tower A, Nanhai Wanda Plaza, No. 28 North Guilan Road, Guicheng Subdistrict, Nanhai District, Foshan City, Guangdong Province, China, to the best of my knowledge of my submitted algorithm LoongKEM and its implementations (reference and optimized implementations included), do hereby declare:

1. I commit that my submitted algorithm and its implementations are in compliance with the security requirements, including but not limited to the requirements that the algorithm does not contain any backdoor or defect artificially designed and its implementations do not contain any malicious code. I do hereby commit that to the best of my knowledge, my submitted algorithm and its implementations do not contain any data, information or technology that may infringe upon trade secrets, or may involve confidentiality required by the laws and regulations of China and other countries.
2. I acknowledge and agree that my submitted algorithm and its implementations will be provided to the public for evaluation, analysis and review. I agree that any interested party, including the program organizer, public reviewers, etc., can freely use the algorithm and its implementations for evaluation purpose. I acknowledge that the program organizer will select several algorithms as candidates for the next-round evaluation or the finalists after evaluation and analysis; my submitted algorithm might not be selected as a candidate for the next-round evaluation or the finalist. I acknowledge that I will not receive financial or other compensation from the program organizer for my submitted algorithm.
3. To facilitate the next-generation commercial cryptographic algorithm standardization, I acknowledge and agree that if my submitted algorithm is selected for the finalists, the program organizer may modify the algorithm and its implementations out of security, availability or other considerations. If my submitted algorithm is selected for the finalists and to be standardized, I commit to place no restriction on the worldwide disclosure and free adoption of algorithm and its implementations.
4. I commit that the algorithm and its implementations are my (or my team's) own original work. The implementations of my submitted algorithm ☐do/☒do not adopt code or code libraries requiring authorization by third parties. Except those clearly identified, citations and acknowledgment, all the opinions, words, diagrams and data in the submission are my own (or my team's) original research results, and the submission does not contain any published work from other individual or team.
5. *(Please check one of the following two options.)*

- ☒ I do not hold and do not intend to hold any patent or patent application which may cover my submitted algorithm and its implementations, and to the best of my knowledge, no patent or patent application may cover my submitted algorithm and its implementations.
- ☐ The following patent(s) and patent application(s) may cover my submitted algorithm and its implementations. As for patent(s) and patent application(s) held by myself, I ☐ *have submitted*/☐ *have not submitted* all statements; as for patent(s) and patent application(s) held by others, I ☐ *have submitted*/☐ *have not submitted* all statements signed by patent and patent application owner(s).

Table The List of Patent(s) and Patent Application(s)

No.	Patent/Patent Application No.	Title of Patent/Patent Application	Patent/Patent Application Owner(s)	Whether submit statements signed by each owner
				☐ <i>Yes</i> ☐ <i>No</i>
				☐ <i>Yes</i> ☐ <i>No</i>

Note: The algorithm submitter(s) shall fill in the table according to his/her/their situation.

I do hereby declare that to the best of my knowledge, I have fully disclosed all patents and patent applications which may cover my submitted algorithm and its implementations. I commit that to the best of my knowledge, my submitted algorithm and its implementations are in compliance with the laws and regulations of China and other countries, and do not infringe upon legitimate rights and interests of any third party. Where there is a dispute concerning patent infringement or violation of laws and regulations, I commit that the algorithm submitter(s) shall take the responsibility of dealing with it and eliminating negative effects as well as bearing corresponding legal liabilities.

I am fully aware that I bear the legal consequences of this statement.

Signed: *Chengdong Tao*

Date: *2026.08.30*

(Note: If there are multiple algorithm submitters, the statements shall be signed by each and every one of them.)

B.2 Statement by Patent (and Patent Application) Owner(s)

(Please check one of the following two options.)

☒ I, Chengdong Tao, Room 2312, Tower A, Nanhai Wanda Plaza,
No. 28 North Guilan Road, Guicheng Subdistrict, Nanhai District, Foshan City,
Guangdong Province, China, am the owner of the patent/patent application
(patent number/patent application number) ;

☐ I, (full name), (full postal address), am the authorized
representative of the owner (full name) of the
patent/patent application (patent number/patent application number)
,

and do hereby agree to grant to any interested party, including the program organizer, public reviewers, etc., the right to freely use the above patent (and patent application) for evaluation purpose, and commit to grant to any interested party, including standard drafter, standard organization department, standard users, etc., a public, irrevocable, nonexclusive, royalty-free license for using the patent (and patent application) without any conditions, if the algorithm LoongKEM is selected for standardization.

Signed: Chengdong Tao

Date: 2026.08.30

(Note: If there are multiple owners of the patent (and patent application), the statements shall be signed by each and every one of them or their authorized representatives. The authorized representative shall submit the letter of authorization.)

B.3 Statement by Reference and Optimized Implementations' Owner(s)

(Please check one of the following two options.)

- ☒ I, Chengdong Tao, Room 2312, TowerA, Nanhai Wanda Plaza, No.28 North Guilan Road, Guicheng Subdistrict, Nanhai District, Foshan City, Guangdong Province, China, am the owner of the submitted reference implementation and optimized implementations of the algorithm LoongKEM;
- ☐ I, (full name), (full postal address), am the authorized representative of the owner (full name) of the submitted reference implementation and optimized implementations of the algorithm (name of algorithm),

and do hereby agree that any interested party, including the program organizer, public reviewers, etc., can freely use such implementations for evaluation purpose, and commit to place no restriction on worldwide disclosure and free adoption if the algorithm is selected for standardization.

Signed: Chengdong Tao
Date: 2026.08.30

(Note: If there are multiple owners of the reference and optimized implementations, the statements shall be signed by each and every one of them or their authorized representatives. The authorized representative shall submit the letter of authorization.)